

Insurance Enrollment Prediction & Agentic Outreach Assistant

1. Business Problem

HR cannot personally contact all ~10,000 employees during enrollment, each region has a fixed `hr_outreach_capacity`. This project predicts which employees are likely to enroll in a new voluntary insurance product, and wraps that model in a small tool-using Outreach Assistant that prioritizes outreach and explains predictions without relying on leaky or protected-attribute information.

2. Data Issues & Cleaning

Two files were profiled and joined: `employees_raw.csv` (10,008 rows) and `region_benefit_profiles.csv` (9 regions, clean, no missing values).

- **Missingness:** `application_date` (719), `last_contact_channel` (1,185), `plan_tier_requested` (526), `broker_channel` (470), `legacy_propensity_score` (908), `outreach_notes` (1,453).
- **Duplicates:** 8 planted `employee_id` conflicts were resolved by sorting on `last_contact_date` and keeping the **most recent contact record** (10,008 → 10,000 rows).
- **Dates:** `application_date` / `last_contact_date` parsed with `errors="coerce"`. Rows where contact occurred *after* application are logically invalid — these `last_contact_date` values were overwritten with `NaT` rather than fed to the model.
- **Text standardization:** `last_contact_channel` (Email/EMAIL/e-mail/Telephone/Text) and `plan_tier_requested` (Premium/premium plan/PREMIUM/Gold Plan) were lowercased, stripped, and mapped to a canonical set.
- **prior_year_enrolled sentinel:** raw values are `-1` (4,048, no prior-year record/new hire), `0` (2,383, not enrolled), `1` (3,577, enrolled). Treated as a 3-way categorical, not a numeric flag. Note: encoding was applied to the raw -1/0/1 codes rather than renamed labels, functionally correct (each group is still separated) but less self-documenting than intended.
- **Region join:** left join on lowercased/striped `region`; all 10,000 rows matched.
- **Not yet investigated:** the brief flags implausibly low `salary` values and `tenure_years` inconsistent with `age`, these were not separately profiled in this iteration.

3. Feature Engineering & Leakage/Fairness Choices

Engineered: `days_since_contact`, `application_month`, `contact_month` (median/mode-imputed where derived from missing dates). **Encoded (one-hot):** `contact`

channel, plan tier, employment type, broker channel, prior_year_enrolled, state_mandate_level.

Category	Fields
Usable	salary, tenure_years, has_dependents, employment_type, broker_channel, plan_tier_requested, engineered date features, region-level fields
Analysis-only	raw date columns, outreach_notes
Forbidden/leaky	employee_id, legacy_propensity_score
Removed for fairness	gender, marital_status, age

legacy_propensity_score (a score from the old rules-based system) was treated with scrutiny precisely because it was suspiciously strong, and dropped from training and from both agent tools, which explicitly refuse to use it.

Not fully mitigated: **hist_enrollment_rate_region** is a same-scope regional aggregate used as-is, without out-of-fold recomputation, a smaller but real leakage risk (flagged, not silently trusted).

Fairness checkpoint: **age**, **gender**, **marital_status** are present and plausibly mildly predictive but were **excluded entirely** from the final model, since using them to prioritize outreach creates disparate-treatment risk. Because the final model contains none of them, no subgroup breakdown applies, per the assignment brief, this is an explicit, justified design choice.

4. Model Training & Evaluation

80:20 **stratified** split (cross-sectional data, not time-series).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (majority class)	0.383	0.000	0.000	0.000	—
Logistic Regression	0.874	0.878	0.924	0.900	0.943
Random Forest (final)	0.913	0.896	0.971	0.932	0.963

Precision@10%: 1.00 — the most business-relevant metric, since HR can only contact a limited number of employees and wants the top of the ranked list reliable. Confusion matrix (RF, n=2,000): 626 TN / 139 FP / 36 FN / 1,199 TP. Random Forest was selected for outperforming Logistic Regression on every metric, including ranking quality. All figures

exclude `legacy_propensity_score` throughout, so they reflect honest, leakage-free performance.

5. Agentic Outreach Assistant

A rule-based, pure-Python router (no LLM) wraps the Random Forest model in four tools plus `assistant()`:

- `predict_enrollment(employee_id)` — model prediction + probability
- `rank_outreach_candidates(region, top_n)` — top-N by probability, defaulting to the region's `hr_outreach_capacity`
- `lookup_region_profile(region)` — region stats
- `explain_prediction(employee_id)` — short explanation citing only operational factors (salary, tenure, prior enrollment, plan tier, contact recency); demographics are never named

Refusal: any query mentioning `legacy_propensity_score` is intercepted before reaching the model and returns an explicit refusal.

Demo transcript :

```
>>> assistant("predict 11")    -> 'Employee not found.'
>>> assistant("region west")   -> region row: enrollment rate 0.613, capacity 437, low
mandate
>>> assistant("predict legacy_propensity_score 1001")
                                -> "Request refused. 'legacy_propensity_score' is a leaky
                                feature and cannot be used for prediction or explanation."
>>> assistant("rank west")     -> 437 employees ranked, top probabilities ≈ 1.00
```

Honest note: `employee_ids` 11 and 1001 aren't in the cleaned dataset, so those two queries returned "not found" rather than a live prediction; refusal and ranking queries worked correctly. Flagged rather than hidden.

6. Future Improvements

- Use valid `employee_ids` in the demo so all five queries return live predictions.
- Investigate the flagged `salary/tenure_years` anomalies.
- Recompute `hist_enrollment_rate_region` out-of-fold to remove residual leakage risk.
- Fix a key mismatch: `predict_enrollment` returns `"probability"` (lowercase) but `explain_prediction` looks up `result["Probability"]` — would raise `KeyError` on a valid employee.
- Add probability calibration, k-fold CV, hyperparameter tuning, and SHAP-based explanations.

7. Conclusion

The pipeline covers leakage-aware cleaning with documented policy decisions, explicit usable/analysis-only/forbidden feature classification, a fairness checkpoint that removes protected attributes outright, honest baseline-vs-model evaluation with the leaky score excluded throughout, and a rule-based agent enforcing both required refusal behaviours. The final Random Forest model (Accuracy 0.913, ROC-AUC 0.963, Precision@10% 1.00) was chosen for its ranking performance, the metric most aligned with the outreach-capacity business problem.